Reflecting Walls Are Self-Collisions: Cellular Automata in a Mirror Box

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October 23, 2026

Code and figures: https://github.com/MiguelPerezI/mirror-walls

Abstract

A cellular automaton with a reflection-symmetric local rule, run on a box whose walls are implemented by mirror ghost cells, is exactly the same automaton on the doubled periodic domain restricted to mirror-symmetric configurations. This is the method of images, and it has a dynamical consequence that we make the organizing principle of the paper: a localized structure that reaches a wall collides head-on with its own mirror image, at the one relative phase the wall admits, and at a stratum of codimension c of the box it meets $2^c - 1$ images. What a wall does to a structure is therefore read off from the bulk collision table, restricted to symmetric collisions. We test this on three automata. In Wolfram's continuous automaton (three-cell mean plus 0.398 modulo 1) the symmetric collision is elastic and the walls are invisible to the population: bulk densities in the box agree with the torus to three digits. In elementary rule 54 we show by exhaustive enumeration that each direction of motion has a single phase class, so every head-on collision of two single movers is mirror-symmetric and annihilates; the wall is a perfect absorber, and because a symmetric state carries as many left as right movers, a sparse box drains all of its movers (in 90 of 90 launches) while a torus keeps the unbalanced excess (in 46 of 90). In rule 110, which is not reflection symmetric, the principle fails and a wall becomes a phase source that can replace the background. In two and three dimensions the boundary strata, indexed by the ternary cube, are quiet in proportion to their codimension, and phase walls are cleared from a layer several cells deep by annihilation with their images. We discuss in what limited sense the box acts as a selecting environment.

1 Introduction

Localized structures of cellular automata (gliders, particles, defects) are usually studied on a ring or a torus, where nothing distinguishes one place from another. Finite domains with reflecting walls are the natural setting of the partial differential equations that many automata discretize, and they are where the question of an *environment* first arises: a wall is a fixed feature that every structure eventually meets. This paper asks what reflecting walls do to localized structures, and answers it with one exact observation and three case studies.

The observation (Section 3) is the method of images: for a rule that is symmetric under $i \mapsto -i$, the box with mirror ghosts is the periodic system on the doubled domain restricted to mirror-symmetric states. It follows that a wall is not a new kind of object for the dynamics. A structure meeting it undergoes the head-on collision with its own mirror image, at the single relative phase that puts the symmetry axis on the wall node. The effect of the wall is

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therefore the outcome of one entry of the bulk collision table, and whether the wall reflects, absorbs or transforms structures is decided by the rule, not by the boundary. Edges and corners of a box are fixed sets of several commuting reflections; we index them by the ternary cube [1], in which each node of an n-dimensional box carries a word in $\{0,1,2\}^n$ recording, per axis, whether it lies on the lower wall, in the interior, or on the upper wall.

The case studies are chosen so that the symmetric collision has three different outcomes. In the one-dimensional continuous automaton of Wolfram [2, ch. 4] at $a = 0.398$ it is elastic, and the walls are invisible (Section 4). In elementary rule 54 [4, 5, 6, 7] it annihilates; we show that in this rule it is the *only* head-on collision of two single movers, so walls and bulk destroy movers by the same event, and the difference between box and torus is a conservation argument: a symmetric state is balanced (Section 5). Rule 110 [8] is not reflection symmetric, and its walls behave as sources (Section 6). In two and three dimensions (Section 7) we measure activity per boundary stratum for the continuous automaton and find quiet strata, quieter with increasing codimension, and a deep boundary layer where the bulk is organized in phase domains.

What is and is not new. The method of images is classical, and the glider collisions of rules 54 and 110 are catalogued [4, 5, 6, 7]. What we add is the reading of reflecting walls as symmetric self-collisions, the balance argument that turns it into a statement about populations, an exhaustive enumeration of the phase classes of the rule-54 movers in the uniform background, and measurements of how boundary strata of every codimension affect a two- and three-dimensional automaton. We found no prior study of localized structures of these automata against reflecting walls.

2 Setting

Continuous automata. On an n-dimensional grid with nodes $\mathbf{i}$ and values $u_{\mathbf{i}} \in [0,1)$, the rule is

$$u_{\mathbf{i}}^{l+1} = \operatorname{frac}\Big(u_{\mathbf{i}}^{l} + \sum_{m=1}^{n} r_m\, \tilde{\delta}^2_{m,\tau_m(\mathbf{i})} u_{\mathbf{i}}^{l} + a\Big), \tag{1}$$

with $r_m = 1/(2n+1)$, that is, the mean of the $(2n+1)$-point neighbourhood plus a constant a, modulo 1. The operator $\tilde{\delta}^2_{m,\tau_m}$ is the per-axis second difference of the ternary-cube rule [1]: on an interior digit $\tau_m = 1$ it is the central difference, on a wall digit it uses the mirror ghost value $u_{-1} := u_1$ (lower wall) or $u_N := u_{N-2}$ (upper wall). This is the explicit heat step with Neumann walls followed by a shift and a fold, a coupled map lattice in the sense of [3]. Wolfram reports class-4 behaviour in one dimension for $a = 0.398$ and $a = 0.4$ [2, p. 244]. Values are rounded to nine decimals after each step to emulate exact arithmetic, which makes runs reproducible across machines.

Elementary automata. For an elementary rule f on $\{0,1\}^N$, the box uses the same mirror ghosts, $u_{-1} := u_1$ and $u_N := u_{N-2}$. A rule is reflection symmetric if $f(\ell, c, r) = f(r, c, \ell)$; rule 54 is, rule 110 is not (its mirror image is rule 124).

Background and defects. Rules 54 and 110 have a periodic background, the ether. For rule 54 it is the tiling by 1000, of temporal period 4; for rule 110 the tiling by 11000100110111, of temporal period 7. Since the ether is known exactly, we display and count structures by comparing the state with the pure ether evolved in the same geometry: a cell is *off-ether* if it differs from it. For rule 54 the mirror box is ether-compatible when its length is $4k+1$, which puts both wall nodes on a 1 of the tile.

3 The reflection principle

Write $\mathcal{R}_m$ for the even reflection of a configuration across the wall nodes of axis m, which turns the box grid $\{0, \dots, N_m - 1\}$ into the periodic grid of length $2(N_m - 1)$, and $\mathcal{R} = \mathcal{R}_1 \cdots \mathcal{R}_n$ for the reflection into the torus $\mathbb{T} = \prod_m \mathbb{Z}/2(N_m - 1)\mathbb{Z}$.

Theorem 3.1 (reflection principle)**.** *Let F be a local rule whose per-axis stencil is symmetric under $i \mapsto -i$, followed by any pointwise map. Let F_{box} be F on the box with mirror ghosts and F_{per} the same rule on $\mathbb{T}$. Then*

$$\mathcal{R} \circ F_{\mathrm{box}} = F_{\mathrm{per}} \circ \mathcal{R},$$

and $\mathcal{R}$ maps the box configurations bijectively onto the configurations of $\mathbb{T}$ that are symmetric under all n reflections, which form an invariant set of F_{per}. This covers (1), its weighted variants [2, p. 244], and every reflection-symmetric elementary rule.

Proof. At a node with an interior digit the two rules read the same neighbours. At a wall node the mirror ghost is exactly the value that the reflected configuration carries at the position beyond the wall, so the stencils agree. Symmetry of the stencil makes the update of a symmetric configuration at $-i$ equal to its update at i, so symmetry is preserved. Pointwise maps commute with both. □

This is the method of images, and for linear stencils it is the reason Neumann problems can be solved by even extension. Numerically the identity holds to the last digit: for (1) on 400 cells for 3000 steps in one dimension and on 64^2 cells for 1000 steps in two dimensions, the maximum difference between the box run and the doubled periodic run is 0 (script `reflection_principle.py`).

Corollary 3.2 (walls are self-collisions)**.** *Under the hypotheses of Theorem 3.1, the evolution of a structure that reaches a wall of the box is the evolution, in the periodic system, of the head-on collision of that structure with its mirror image, at the relative phase that puts the symmetry axis on the wall node. At a node of codimension c (a word of the ternary cube with c wall digits) the structure meets $2^c - 1$ images under c commuting reflections. A configuration supported on a stratum is a configuration of the periodic system that is symmetric under the corresponding reflections.*

Two consequences are used below. First, the wall realizes only the *symmetric* entries of the collision table, and it realizes them for every structure that reaches it: if a rule has several head-on phases with different outcomes, the wall picks out one, always the same. Second, a stratum is quiet for a kinematic reason: the wall node is a symmetry centre, so only mirror-symmetric local patterns can sit on it, and a node of codimension c has c of its neighbour pairs identified, so it has fewer independent neighbours to differ from.

Proposition 3.3 (balance)**.** *Let a reflection-symmetric one-dimensional rule have, in a background, movers of speed $+v$ and $-v$ that are mirror images of each other.*

1. *In the doubled system of a box, the number of right movers equals the number of left movers at all times.*

2. *Suppose that in some regime movers are neither created nor converted, that two movers of the same direction never meet, and that every head-on meeting of a right and a left mover annihilates both. Then on a ring the number $N_+ - N_-$ is conserved and $|N_+ - N_-|$ movers survive forever, while in the box every mover is eventually annihilated.*

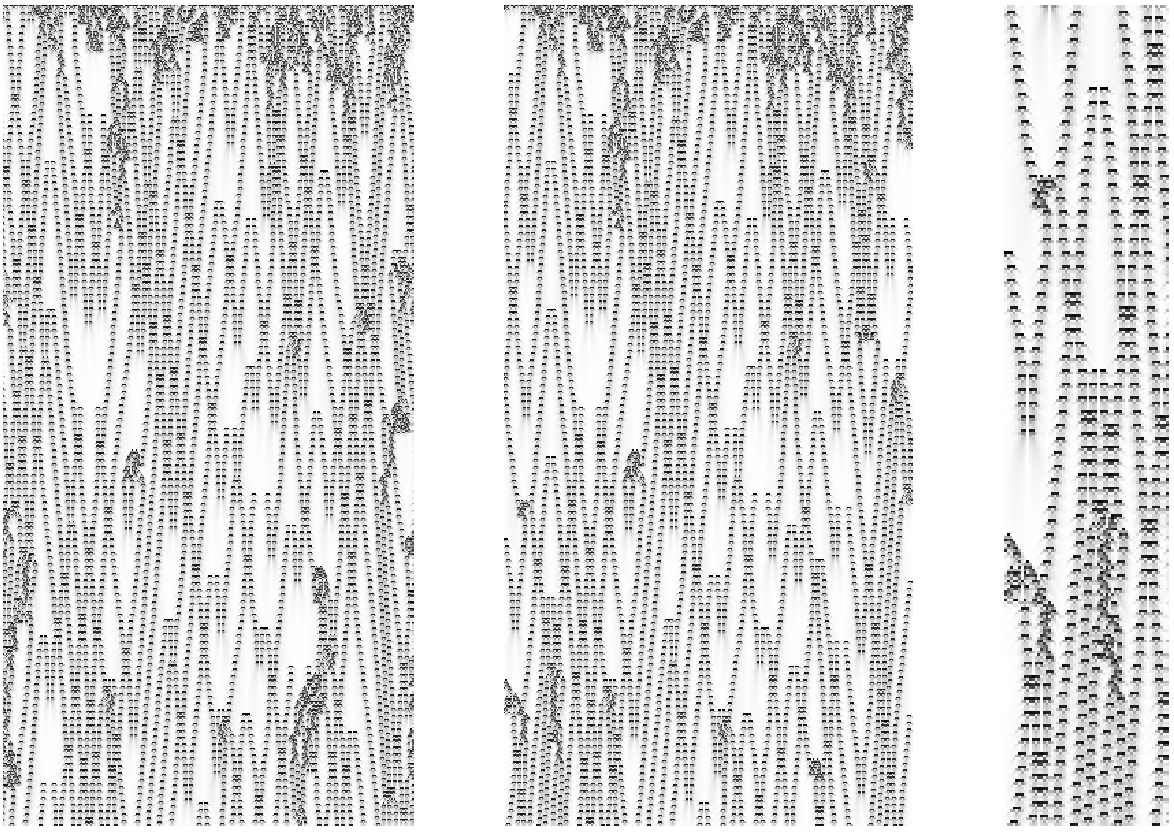


Figure 1: Class 4 in one dimension: $a = 0.398$, three-cell mean plus a modulo 1, 300 cells, 600 steps downward from random data, same seed, difference display (black where neighbours differ). Left: ring. Middle: mirror walls at both sides. Right: the 60 cells nearest the left wall during steps 300–600, magnified. Localized structures move through a periodic background and at the wall return as their mirror images, forming V shapes.

Proof. (1) The reflection maps right movers to left movers and the doubled state is symmetric. (2) Each annihilation removes one mover of each direction, which conserves $N_+ - N_-$. On a ring of length L a right and a left mover approach at relative speed $2v$ and meet within $L/(2v)$ steps, so movers are annihilated in pairs until only one direction remains, and movers of one direction never meet. In the box, the doubled ring has $N_+ = N_-$ by (1), so all movers are annihilated. □

4 Elastic walls: the continuous automaton

We take (1) in one dimension with $a = 0.398$ from random data and the activity indicator $\chi_i = [\mathrm{dist}(u_i - u_{i+1}, \mathbb{Z}) > 0.05]$, which is Wolfram's difference display thresholded. Figure 1 shows the space-time diagrams on a ring and in a box.

The zoo. Tracking clusters of active cells on a 600-cell ring over 3000 steps gives one dominant mover, of speed 0.1 cells per step in either direction and width about 3.5 cells, and a rare slow, wider class (about 5% of tracks), which is born in collisions and lives about 60 steps against about 105 for the movers. A head-on collision of two movers returns the pair, so by Corollary 3.2 a mover hitting a wall should come back, and it does (Figure 1, right).

The walls are invisible to the population. Over ten seeds, 400 cells and 10^4 steps after a transient, counting active clusters in the bulk (cells 50 to 350) at every step, the ring and the box give 0.0723 clusters per cell per step both, and fractions 0.406 and 0.408 of wide clusters (width at least 6). Mean lifetimes of both classes agree between the geometries (104 against 105 steps for movers, 59 against 60 for the slow class). Total cluster counts over $2 \cdot 10^4$ steps fluctuate around 30 in both geometries with no trend. Near the walls, the wall node has activity 0.09 and its neighbour 0.23 against a bulk value of 0.27, after which the bulk value is reached: a quiet layer two cells deep, as Corollary 3.2 predicts, and nothing else. Walls that reflect elastically do not select.

5 Absorbing walls: rule 54

Rule 54 has the ether 1000, movers $w_\pm$ of speed ± 1, stationary breathers, and glider guns [4, 5, 6, 7]. Its head-on collisions are known to depend on the relative phase: gliders "in the same phase" annihilate, gliders in anti-phase fuse into a stationary breather [6].

One phase class per direction. We insert every binary word of length up to 10 at each of the four offsets into the ether, evolve 160 steps on a ring, and keep the outcomes that consist of a single mover, identified as off-ether cells moving rigidly at speed one (script `rule54_exact.py`). Up to translation by the ether period, exactly one shape is found for each direction: a pair of adjacent off-ether cells, starting at residue 2 modulo 4 (right movers) or 1 modulo 4 (left movers) at times divisible by 4; the other outcomes are trains of two or more movers four cells apart. A survey of every mover in 1500 random launches of 20 cells (script `rule54_phases.py`), 3×10^5 observations including movers born in multi-body events, finds no other residue. So in the uniform ether the phase of a mover is not a free parameter, and the relative phase of a head-on collision of two single movers is fixed.

The collision and the wall event. Placing a single right and a single left mover at every admissible pair of offsets on a ring, we find one collision class: it is mirror-symmetric about a node at all times and annihilates both movers, leaving pure ether. This is the "same phase" annihilation of [6]; the anti-phase fusion does not arise between single movers in the uniform ether. By Corollary 3.2 the wall event is the same collision, and directly: a single right mover started at any offset in a box of length $4k + 1$ meets the wall in one way, and nothing is left. A train of four movers meeting a single mover loses one member and continues; trains of two and three meeting the wall lose all their members (Figure 2).

Sparse populations. One random 8-cell patch is placed at the centre of the ether, in a ring of 400 cells and in a box of 401 cells with the same patch, and the movers are counted exactly (a right mover is a pair of off-ether cells at residue 2 that reappears four cells to the right four steps later; script `rule54_sparse.py`). Of 200 patches, 110 produce a glider gun, which fills either geometry with movers at the same rate; Table 1 summarizes the other 90.

This is Proposition 3.3 at work. The hypotheses hold in the sparse regime by the enumeration above: every head-on meeting of single movers annihilates, and trains lose one member per meeting. On the ring the unbalanced movers never meet a partner and live forever; in the box each mover's partner is its own image, so all of them die, each within one crossing of the box (Figure 3). In the two exceptional rings the patch leaves a small stationary structure behind, and the single mover is absorbed by it on its first pass, an interaction outside the hypotheses of the proposition (stationary structures can absorb, reflect or convert movers [6]).

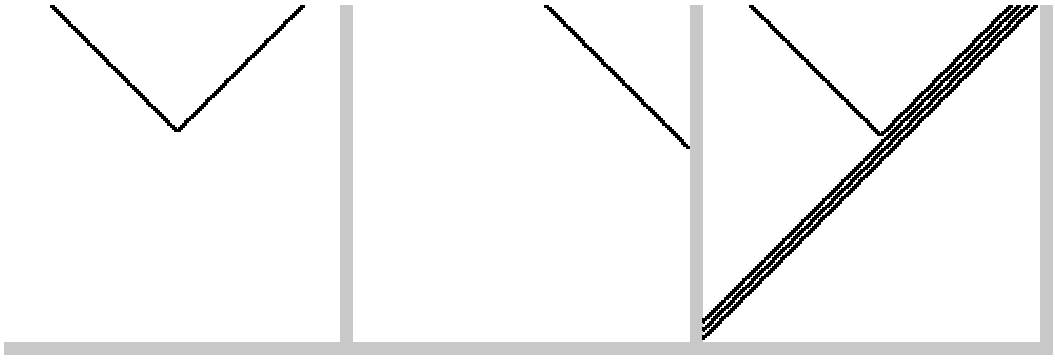


Figure 2: Rule 54, off-ether display (black where the state differs from the exact ether), time downward, 160 steps by 160 cells. Left: a single right and a single left mover meet on the ring and annihilate. Middle: a single right mover reaches the mirror wall (right edge of the panel) and vanishes: the same event, with the left mover supplied by the reflection. Right: a single right mover meets a train of four left movers; one pair annihilates and three continue.

	mean number of movers at step					
geometry	100	300	600	1000	1500	2000
ring (400 cells)	2.06	1.17	1.14	1.14	1.14	1.14
box (401 cells)	2.06	0.00	0.00	0.00	0.00	0.00

Table 1: Rule 54, 90 sparse launches without guns, same patches in both geometries. At step 2000 movers survive in 46 of the 90 rings and in none of the boxes. In every ring the survivors move in a single direction, and $N_+ - N_-$ at step 300 equals $N_+ - N_-$ at step 2000 in 88 of the 90 rings.

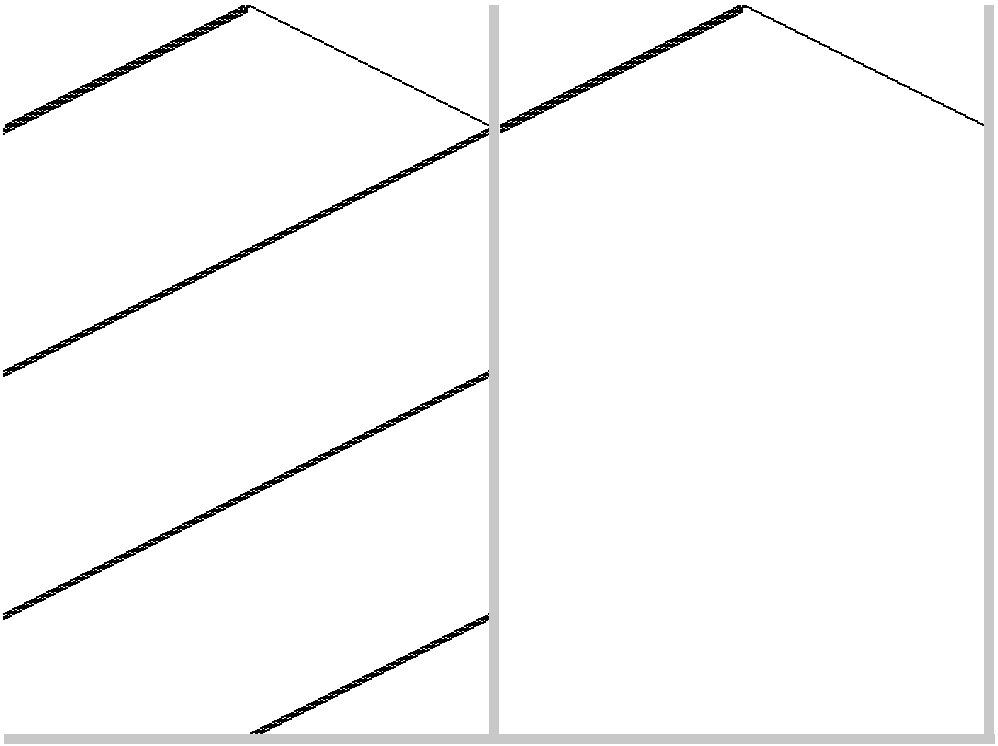


Figure 3: One sparse launch, off-ether display, 1200 steps (every second step shown), ring of 400 cells (left) and box of 401 cells (right), same patch. The patch emits a train of three left movers and one right mover. On the ring the right mover meets the train across the periodic seam and annihilates one member; the other two circulate forever. In the box every mover meets its image at a wall.

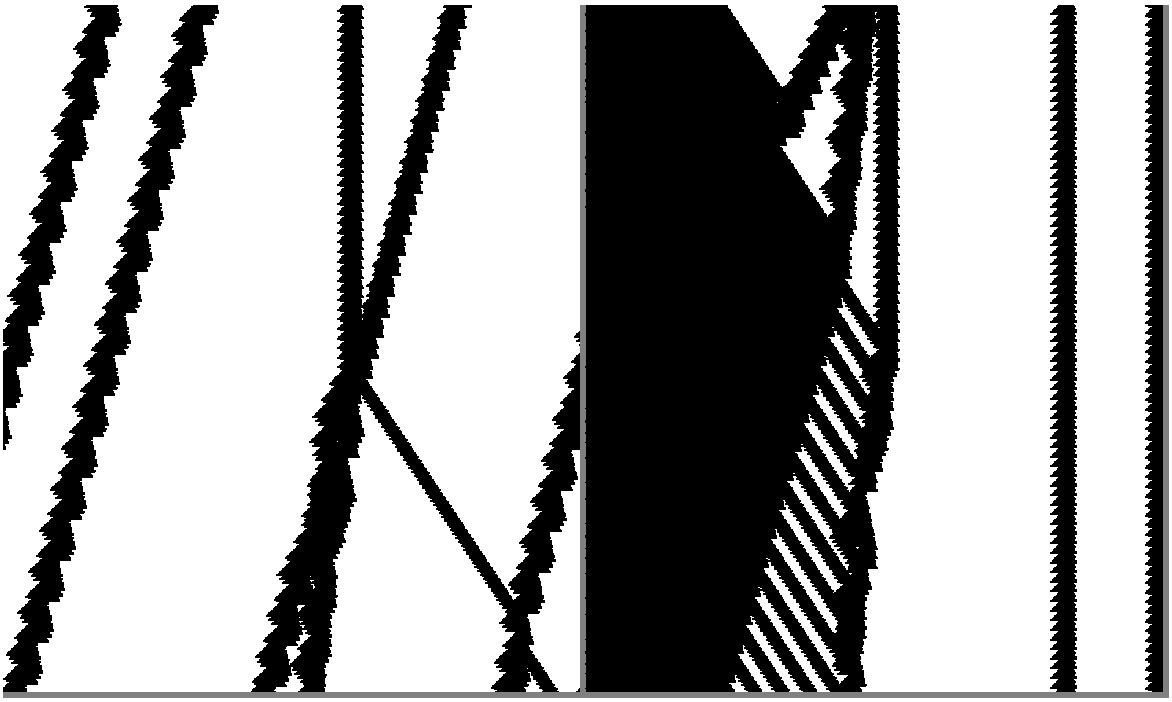


Figure 4: Rule 110, defect display (black where the 14-cell word is not an ether word), time downward. Left: ring. Right: box with mirror walls, same initial data; the left wall has nucleated a different background, which advances into the box and emits gliders.

Dense populations. From random data on 1200 cells the population is sustained by guns, and walls and bulk both destroy movers by the same collision. Over three seeds and six sampling times between steps 1000 and 6000 the ring carries 107 ± 9 movers and the box 94 ± 13: no difference beyond the fluctuations, which are dominated by the birth and death of guns. The walls are perfect absorbers, but in a dense population the bulk is nearly as effective.

6 Asymmetric rules: rule 110

Rule 110 is not reflection symmetric, so Theorem 3.1 does not apply and a mirror wall is a genuinely new boundary rule, not a self-collision. Its ether has spatial period 14 and is not mirror-compatible: the right wall, which is nearly ether-compatible, hosts stationary structures and little else, but the left wall can be kicked by an arriving glider into a state that grows a different periodic background. That background invades the box behind a front, emitting fast right-moving gliders as it advances (Figure 4). Once nucleated the invasion is total: in one of two 1400-cell runs the ether fraction fell from 0.90 at step 5000 to 0 at step 8000; in the other it stayed at 0.96 through step 8000. The wall changes the population by emission and eventually destroys the medium the gliders live in. Without the reflection symmetry there is no principle that predicts what a wall does, and the outcome depends on the details of the boundary rule.

7 Two and three dimensions

Regimes. In two dimensions (1) is the five-point mean plus a modulo 1. A scan of a in steps of 0.01 on a 128^2 torus from random data (script `scan2d.py`) gives a late activity fraction that

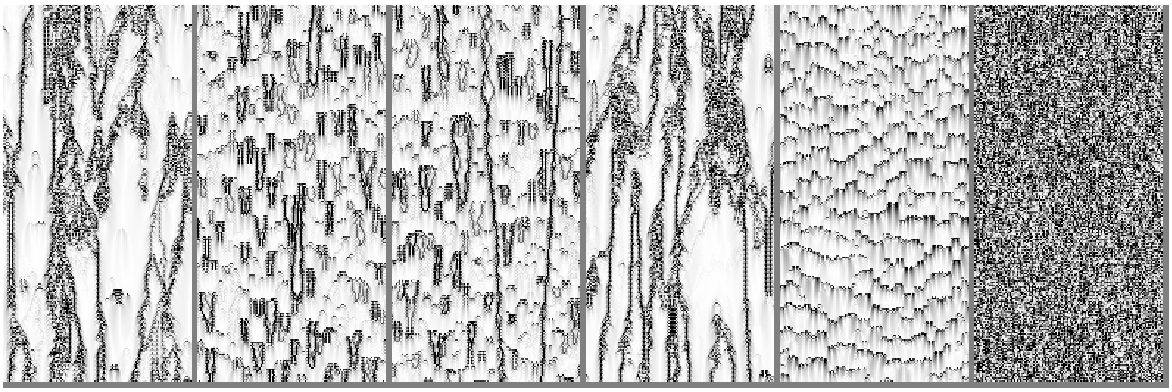


Figure 5: Space-time cuts (middle row of a 128^2 torus against the last 256 of 1200 steps, difference display) for $a = 0.33$, 0.49, 0.51, 0.67, 0.02, 0.398. Wandering phase walls at 0.33 and 0.67, intermittent spots at 0.49 and 0.51, a regular lattice at 0.02, chaos at 0.398.

regime	grid	codim 0	codim 1	codim 2	codim 3
chaos, $a = 0.398$	128^2	0.989(0)	0.948(0)	0.738(3)	
chaos, $a = 0.398$	32^3	0.997(0)	0.993(0)	0.981(0)	0.922(3)
intermittent, $a = 0.49$	128^2	0.558(2)	0.494(2)	0.196(4)	
intermittent, $a = 0.49$	32^3	0.484(5)	0.505(5)	0.506(3)	0.305(8)
phase walls, $a = 0.33$	128^2	0.510(5)	0.340(3)	0.172(11)	
chaos (3-D rule), $a = 0.33$	32^3	0.996(2)	0.992(5)	0.973(7)	0.860(10)

Table 2: Time-averaged activity fraction per boundary stratum of the box with mirror walls, mean over eight seeds with the standard error in the last digits in parentheses.

varies continuously with a, from 0.29 near $a = 0.01$ to 0.92 on the plateaus $a \in [0.34, 0.43]$ and $[0.57, 0.66]$, with dips to 0.43–0.45 near $a = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}$. There are no sharp class-4 windows of the one-dimensional kind, and the one-dimensional class-4 value $a = 0.398$ is fully chaotic in two dimensions. Space-time cuts (Figure 5) show three regimes: near $a = \frac{1}{3}$ and $\frac{2}{3}$ the background is a period-three cycle and the dynamics consists of slowly wandering *phase walls* between its three phases; near $a = \frac{1}{2}$ there are intermittent short-lived spots on a period-two background, in the manner of spatiotemporal intermittency [9]; on the plateaus, spatiotemporal chaos. We found no gliders in this family in two dimensions; its localized structures are the phase walls and the spots.

Activity per stratum. We run the box with mirror walls, classify every node by the codimension of its stratum (0 interior, 1 face, 2 edge or two-dimensional corner, 3 three-dimensional corner), and record the time-averaged activity indicator, defined as the maximum over all existing neighbour differences of $\operatorname{dist}(u_{\mathbf{i}} - u_{\mathbf{j}}, \mathbb{Z}) > 0.05$. Grids are 128^2 for 1500 steps and 32^3 for 800 steps, averaged after a transient of one third of the run, over eight seeds (script `strata_seeds.py`). Table 2 gives the results.

Four observations. First, in the chaotic regime the activity decreases monotonically with the codimension of the stratum, in both dimensions, and the boundary layer is one cell deep: at distance one from the nearest wall the two-dimensional activity is already 0.982 against 0.989 in the bulk. This is the kinematic effect of Corollary 3.2: a node of codimension c has c of its neighbour pairs identified by symmetry, so it has fewer independent neighbours to differ from, and the effect compounds with c. Part of the trend is therefore a property of the indicator, which takes a maximum over fewer independent differences on the strata. Second,

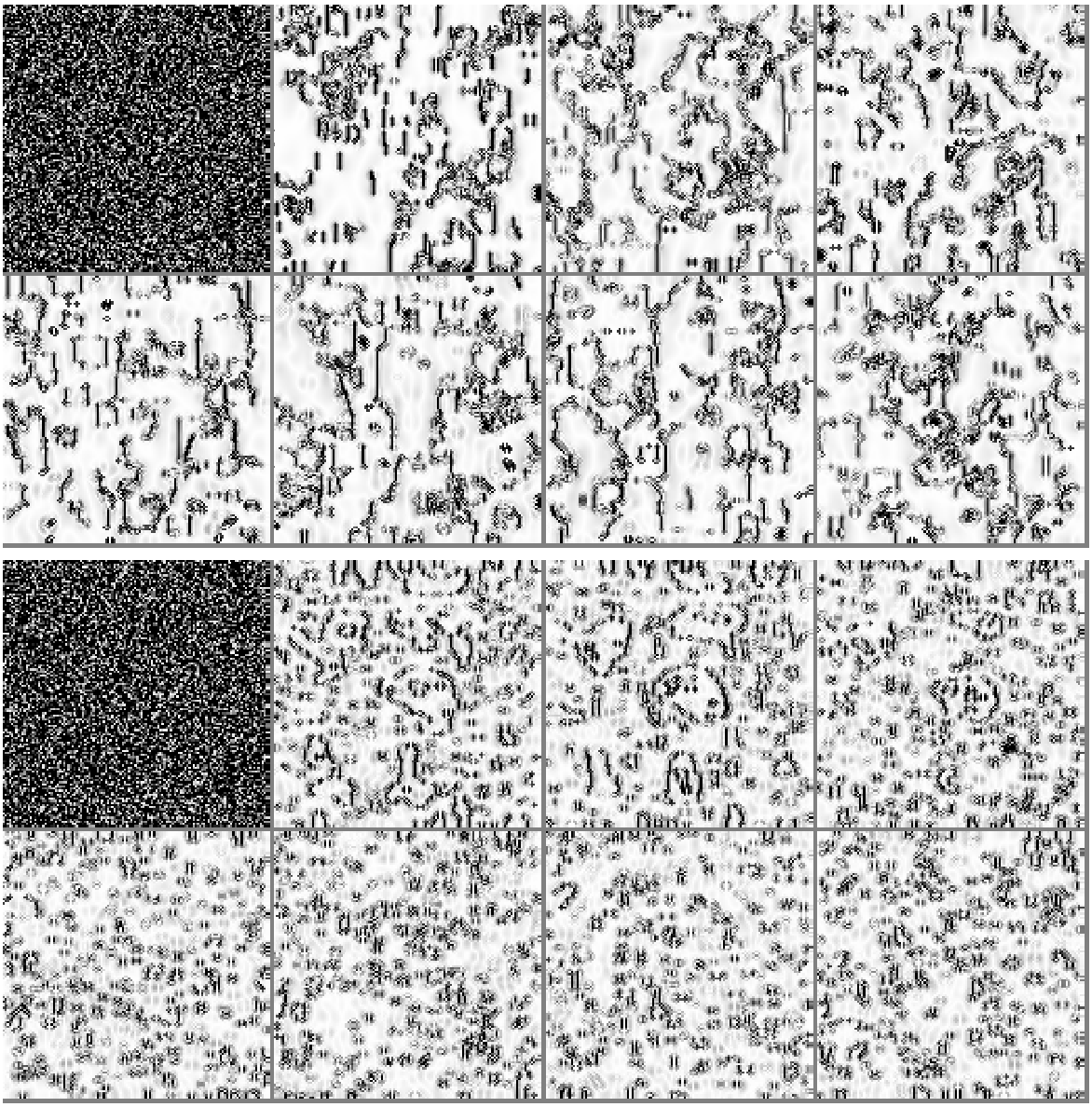


Figure 6: The 128^2 box with mirror walls, difference display, at steps $0, 50, 100, 200$ (top row of each strip) and $400, 600, 800, 1200$ (bottom row). Upper strip, $a = 0.33$: random data collapse within fifty steps into a period-three background crossed by phase walls, which then wander, join and pinch off; walls parallel to a side are cleared from a margin several cells wide, walls perpendicular to it end on it. Lower strip, $a = 0.49$: intermittent spots on a period-two background, born and dying everywhere except at the corners.

in the intermittent regime the corner is strongly suppressed in both dimensions (0.20 against 0.56 in two dimensions, 0.31 against 0.48 in three), while the three-dimensional faces and edges sit slightly above the bulk value. So in two dimensions the spots avoid the corners but not the walls, and in three dimensions only the eight corner nodes are special; the twelve edges, which are the analogue of the two-dimensional corners, are not. Third, the phase-wall regime is the one in which the box is felt at a distance: the activity profile against the distance to the nearest wall is 0.34, 0.38, 0.43, 0.45, 0.46, 0.47 for distances 0 to 5, still rising at distance five, against one cell in the other regimes. Corollary 3.2 says why: a phase wall parallel to a mirror meets its image, a phase wall of opposite orientation, and the pair annihilates when the strip between them collapses, so a margin along each mirror is cleared of parallel phase walls; phase walls perpendicular to the mirror are their own images, end on it, and are the activity that remains at distance zero. This is the two-dimensional counterpart of the absorbing walls of rule 54. Fourth, the same $a = 0.33$ is chaotic under the seven-point rule in three dimensions, with the one-cell layer of the chaotic regime, which shows that the regime, not the constant, is what the strata respond to.

8 Discussion: is the box a selecting environment?

Wong et al. [11] have proposed that the functional information of a system, in the sense of [12, 13], increases when many configurations are subjected to selection for some function, and have asked which environments act as selectors. A box with mirror walls is about the simplest fixed environment a cellular automaton can have, and the results above say precisely what it selects. The "function" is surviving one's own reflection: the wall applies to every structure that reaches it the symmetric entry of the collision table, and keeps exactly the structures for which that entry is elastic. In the continuous automaton of Section 4 every structure passes and the box selects nothing. In rule 54 no mover passes, so the box filters a sparse population to its stationary members, while the ring keeps a conserved excess of movers; this is differential persistence imposed by the environment, and it is the only regime in which we see the box change a population. It is one-shot filtering toward a fixed survivor set, not the open-ended accumulation the law of [11] describes: nothing is copied, and nothing varies. In rule 110 the walls do not filter but emit.

Selection among structures, as opposed to against all of one kind, requires a reflection-symmetric rule with several mover species whose symmetric self-collisions differ, some elastic and some not. Rule 54 has the symmetry but one mover species; rule 110 has a rich zoo [8] but not the symmetry. Two-dimensional continuous automata of Lenia type [10] have both, and add the corner, a four-body symmetric collision, as a second and independent filter. The experiment of this paper (exhaustive wall events, sparse launches with the same seeds in both geometries, exact counting) transfers to them unchanged.

Code

All scripts, the figures and the source of this article are public at `https://github.com/MiguelPerezI/mirror-walls`. In `code/`: `contca.py` (rule (1) with ternary-cube walls) and `reflection_principle.py` (Theorem 3.1); `view1d.py`, `tracker.py`, `perstep.py` and `population.py` (Section 4); `eca.py`, `rule54_exact.py`, `rule54_phases.py`, `rule54_sparse.py`, `rule54_charge.py` and `fig_rule54.py` (Section 5); `eca_box.py` (Section 6); `scan2d.py`, `cuts2d.py`, `filmstrip2d.py` and `strata_seeds.py` (Section 7). `view2d.py` is a live viewer of the two-dimensional box (`python3.12 view2d.py 0.33 160 mirror`).

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